

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING (C-DAC), MUMBAI
C++ PROGRAMMING
Question Paper — PGCP-AC A26

Date: 06/09/2026	Duration: 120 Minutes	Total Marks: 40
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QUESTION 1: Student Grade Management System (20 Marks)

You are designing a grade management module for a university. The system needs to store student records and calculate grades efficiently.

Design a class Student with the following:

Private Data Members:

- int rollNumber — unique student ID
- string name — student name
- double percentage — marks obtained as percentage (0-100)
- static int totalStudents — count of total Student objects created

Public Member Functions:

1. Parameterized Constructor:

```
Student(int roll, const string& studentName);
```

- Initialize percentage to 0
- Increment totalStudents
- Print: "[Student Created] Roll: <roll>, Name: <name>"

2. Copy Constructor — Implement copy constructor

- Print: "[Copy Constructor Called]"

3. Destructor

- Decrement totalStudents
- Print: "[Student Deleted] Roll: <roll>"

4. Setter and Getter:

```
void setPercentage(double percent); // Validate:  $0 \leq \text{percent} \leq 100$   
double getPercentage() const;
```

5. Grade Calculation:

```
char calculateGrade() const; // Return A ( $\geq 80$ ), B ( $\geq 70$ ), C ( $\geq 60$ ), D ( $\geq 50$ ), F  
(<50)
```

6. Operator Overloads:

Stream Operator *operator<<* (*friend function*) — Print student details in format:

```
Roll: 101 | Name: Raj Kumar | Percentage: 86.6% | Grade: A
```

Equality Operator *operator==* — Compare students by roll number

7. Utility Functions:

```
static int getTotalStudents(); // Return totalStudents
```

main() Function Requirements:

- Use vector<Student> to store multiple Student objects
- Create at least 5 Student objects and add them to the vector
- Set percentage for each student (use setPercentage())
- Print all students using cout << student or loop through vector
- Demonstrate copy constructor by copying one student
- Use operator== to verify roll numbers match between two students
- Find and display student with highest percentage
- Find and display student with lowest percentage
- Print total students count at different stages
- Show grade distribution (count students in each grade A, B, C, D, F)

Expected Output (partial example):

```
[Student Created] Roll: 101, Name: Raj Kumar
[Student Created] Roll: 102, Name: Priya Singh
[Student Created] Roll: 103, Name: Arun Patel
[Student Created] Roll: 104, Name: Deepa Sharma
[Student Created] Roll: 105, Name: Vikram Singh
Roll: 101 | Name: Raj Kumar | Percentage: 86.6% | Grade: A
Roll: 102 | Name: Priya Singh | Percentage: 90.8% | Grade: A
Roll: 103 | Name: Arun Patel | Percentage: 70.0% | Grade: B
Roll: 104 | Name: Deepa Sharma | Percentage: 80.0% | Grade: A
Roll: 105 | Name: Vikram Singh | Percentage: 55.0% | Grade: D
Total Students: 5
Highest Percentage: Priya Singh (90.8%)
Lowest Percentage: Vikram Singh (55.0%)
Grade Distribution:
Grade A: 3 students
Grade B: 1 student
Grade D: 1 student
[Copy Constructor Called]
Total Students after copy: 6
[Student Deleted] Roll: 101
[Student Deleted] Roll: 102
[Student Deleted] Roll: 103
[Student Deleted] Roll: 104
[Student Deleted] Roll: 105
[Student Deleted] Roll: 101 // ← Copied student has same roll number as original
```

QUESTION 2: Employee Bonus Calculator (10 Marks)

Design a class Employee that calculates bonus based on performance rating and salary.

Class Requirements:

Private Members:

- int empId
- string name
- double salary
- int performanceRating (1-5 scale)

Public Functions:

1. Constructor:

```
Employee(int id, const string& empName, double sal, int rating);
```

2. Calculate Bonus:

```
double calculateBonus() const;
```

- Rating 5: 20% bonus
- Rating 4: 15% bonus
- Rating 3: 10% bonus
- Rating 2: 5% bonus
- Rating 1: 0% bonus

3. Display Function:

```
void displayDetails() const;
```

Print format: ID: 001 | Name: Raj Kumar | Salary: 50000 | Rating: 4 | Bonus: 7500

main() Implementation:

- Create 5 Employee objects using array
- Set different ratings for each
- Calculate and display bonus for each employee
- Find and display the employee with maximum bonus

Expected Output:

```
ID: 001 | Name: Raj Kumar | Salary: 50000 | Rating: 4 | Bonus: 7500
ID: 002 | Name: Priya Singh | Salary: 60000 | Rating: 5 | Bonus: 12000
ID: 003 | Name: Arun Patel | Salary: 45000 | Rating: 3 | Bonus: 4500
ID: 004 | Name: Deepa Sharma | Salary: 55000 | Rating: 4 | Bonus: 8250
ID: 005 | Name: Vikram Singh | Salary: 52000 | Rating: 2 | Bonus: 2600
Employee with Maximum Bonus: Priya Singh with Bonus: 12000
```

QUESTION 3: Temperature Converter (10 Marks)

Write a program that converts temperatures between Celsius, Fahrenheit, and Kelvin.

Requirements:

1. Create a struct/class Temperature with:

- double celsius
- Constructor accepting temperature value
- Three conversion functions:

```
double toFahrenheit() const; //  $(C \times 9/5) + 32$   
double toKelvin() const; //  $C + 273.15$   
void displayAll() const; // Display all three scales
```

2. Validate input:

- Celsius must be ≥ -273.15 (absolute zero)
- Print error if invalid

3. main() Function:

- Accept temperature in Celsius from user
- Validate input
- Display temperature in all three scales
- Process at least 3 temperature values (use loop or multiple inputs)

Expected Output:

```
Enter temperature in Celsius: 25  
Temperature: 25°C = 77°F = 298.15K  
Enter temperature in Celsius: 0  
Temperature: 0°C = 32°F = 273.15K  
Enter temperature in Celsius: 100  
Temperature: 100°C = 212°F = 373.15K  
Enter temperature in Celsius: -300  
ERROR: Temperature below absolute zero (-273.15°C)
```

— END OF QUESTION PAPER —